

# Scale Factor Confusion: Linear, Square, Cubic

## Answer Key

### High School Geometry

#### Quick Answers

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- |          |                                |
|----------|--------------------------------|
| 1. 108   | 4. $\frac{5}{3}, -\frac{5}{3}$ |
| 2. 432   | 5. 2                           |
| 3. 112.5 | 6. —                           |
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#### Common wrong answers

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- 1. *If they got 36: multiplied the area by the linear scale factor instead of by its square*
  - 2. *If they got 108: multiplied the volume by the linear scale factor instead of by its cube*
  - 2. *If they got 216: used the square of the factor, the rule for areas, on a volume*
  - 3. *If they got 45: scaled the area by the linear ratio instead of by its square*
  - 5. *If they got 8: reported the volume ratio itself as the linear scale factor*
  - 5. *If they got 2.83: took the square root of the volume ratio, the rule for areas*
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**The single idea being tested:** the exponent on the scale factor counts *dimensions*. Lengths scale by  $k$ , areas by  $k^2$ , volumes by  $k^3$ ; running the rule backwards takes the matching root. Every wrong answer on this sheet is the right factor raised to the wrong power.

#### 1. Area 12 after a dilation with $k = 3$ .

A dilation multiplies every length by  $k$ , and an area is built from two perpendicular lengths, so both of them stretch: the area is multiplied by  $k \cdot k = k^2$ .

$$A' = 12 \cdot 3^2 = 12 \cdot 9$$

So the image has area  $108 \text{ cm}^2$ .

*Watch for:*  $12 \cdot 3 = 36$ , which scales the area as if it were a length.

#### 2. Similar cones, linear factor 2, smaller volume 54.

A volume is built from three perpendicular lengths, and a dilation stretches all three, so the volume is multiplied by  $k \cdot k \cdot k = k^3$ . With  $k = 2$ , that factor is 8.

$$V' = 54 \cdot 2^3 = 54 \cdot 8$$

The larger cone holds  $432 \text{ cm}^3$ .

*Watch for:* 108 (the factor used once, as if volume were a length) and 216 (the factor squared, which is the rule for areas).

**3. Find the mistake: similar triangles, 5 : 2, smaller area 18.**

(a) Rey used the *linear* ratio on an *area*. The ratio 5 : 2 compares corresponding sides, so it scales lengths. Area needs the square of that ratio, because both dimensions of the triangle grow.

(b) Correct work:  $k = \frac{5}{2}$ , so the area ratio is  $k^2 = \frac{25}{4}$ , and

$$A_{\text{larger}} = 18 \cdot \left(\frac{5}{2}\right)^2 = 18 \cdot \frac{25}{4} = \frac{450}{4}$$

which gives  $\boxed{112.5 \text{ cm}^2}$ .

*Sanity check:* the larger triangle is more than twice as long in each direction, so more than a doubling of area is expected — and 112.5 is more than six times 18, while Rey's 45 was only two and a half times.

**4. Areas in the ratio 25 : 9: solve  $9k^2 = 25$ .**

(a) The equation says the area ratio is the *square* of the linear factor, so isolate  $k^2$  and take roots. Squaring destroys sign information, so the equation has two solutions:

$$k^2 = \frac{25}{9} \quad k = \pm \sqrt{\frac{25}{9}}$$

which are  $\boxed{k = \frac{5}{3}}$  and  $\boxed{k = -\frac{5}{3}}$ .

(b) A scale factor multiplies lengths, and lengths are positive, so only  $k = \frac{5}{3}$  describes a pair of similar figures; the negative root is an artifact of squaring and is rejected. As a check,  $\left(\frac{5}{3}\right)^2 = \frac{25}{9}$ , the given area ratio.

**5. Find the mistake: similar cylinders of volume 250 and 2000.**

(a) Nia reported the *volume* ratio as if it were the linear factor:  $2000/250 = 8$  is  $k^3$ , not  $k$ . Kim took a square root, which is the rule for going back from an *area* ratio; on a volume ratio it gives about 2.83, a number that is neither the linear factor nor anything else in the problem.

(b) Volume ratio is  $k^3$ , so the linear factor is the *cube* root:

$$k^3 = \frac{2000}{250} = 8 \quad k = \sqrt[3]{8}$$

The correct linear scale factor is  $\boxed{k = 2}$ .

*Check it forwards:*  $250 \cdot 2^3 = 2000$ , which is exactly the larger volume, so the factor is confirmed — a check neither student ran.

**6. “Double every edge: twice the paint, twice the sand.”** (open response — read the sketch and the reasoning)

The claim is wrong in both halves, and by different amounts. With every edge doubled the linear scale factor is  $k = 2$ . Paint covers the outside, which is surface area, so it scales by  $k^2 = 4$ : the box needs four times as much paint. Sand fills the inside, which is volume, so it scales by  $k^3 = 8$ : it holds eight times as much sand. Only a *length* — an edge, a diagonal, the perimeter of a face — doubles.

|                                    |
|------------------------------------|
| paint $\times 4$ , sand $\times 8$ |
|------------------------------------|

*Full credit* shows a sketch of the original box and the doubled box with edges labelled, states  $k^2$  for surface area and  $k^3$  for volume, and explains that the exponent counts how many dimensions are being stretched at once (a face is stretched in two directions, the interior in three). *Not full credit*: the numbers 4 and 8 with no link to dimensions, or a claim that only the volume changes.